

# Cognitive Operator Theory:

A Curvature Dependent Model of Organizational Cognition

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## Abstract

This paper explains how an organization's structure shapes its ability to think, learn, coordinate, and make decisions. Building on earlier work that showed how centralization can be measured as "curvature" in the organization's communication network, the current paper develops a full computational model of how organizations process information. The model introduces three core mechanisms. The *cognition operator* describes how information is amplified or lost as it moves through the organization. The *decision operator* explains how groups convert information into stable choices. The *cognitive load operator* measures how much strain the organization experiences when processing information.

Four findings matter directly for leaders. First, highly centralized structures dramatically reduce the organization's ability to amplify and integrate information. Second, decentralized structures make decision making more stable and resilient. Third, the organization's "cognitive capacity" its ability to handle complexity declines predictably as communication bottlenecks increase. Fourth, coordination capacity and cognitive capacity are two sides of the same coin: improving one automatically improves the other.

The paper also provides a measurement framework that allows leaders to estimate these quantities directly from communication data (e.g., email or messaging logs). A simple four-node example shows how small structural differences can produce large differences in cognitive performance. Overall, the paper offers a unified, geometry-based explanation of how organizational design shapes thinking, decision making, and the ability to manage cognitive load.

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# 1 Technical Introduction

This paper develops a mathematically rigorous, operator theoretic framework for organizational cognition, extending the geometric foundation established in *Geometry of Decentralization* [18]. That work identified the Riemannian curvature  $\mathcal{R}_{\mathcal{K}}$  of the information-flow manifold  $(M, g_{\mathcal{K}})$  as a geometric invariant of centralization, with the Ricci spectral gap  $\Delta(\kappa)$  governing the transition between mechanistic, organic, and DSMS regimes. Building on this structure, the present paper introduces three operators that formalize the computational dynamics of organizations: a nonlinear *cognition operator*  $A$  that amplifies and transforms information; a *decision operator*  $\Delta$  that produces stable choices as fixed points of cognitive processing; and a *cognitive load operator*  $L$  that quantifies the burden imposed by amplified information.

Four main results are established with complete proofs. *First*, the curvature-dependent amplification law,

$$\|DA(x^*)\| = \Theta(e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}),$$

is derived under an explicit non-degeneracy condition, with  $\|\mathcal{R}_{\mathcal{K}}\|$  operationalized via the between concentration metric of [18]. *Second*, decision stability is shown to arise from curvature induced contraction of the composite map  $F = \Delta \circ A$ . *Third*, the dominant eigenvalue of the linearized amplification operator satisfies  $\lambda_1 \asymp e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}$ , ensuring spectral consistency with the amplification law. *Fourth*, amplification and coordination bandwidth are proven spectrally dual, linking cognitive capacity directly to coordination geometry.

$$\|DA(x^*)\| \asymp \|B_K\|,$$

A complete measurement framework connects each operator to observable organizational quantities, and a worked example on four-node communication structures

illustrates the theory concretely. Together, these results yield a unified geometric account of how organizational structure shapes cognition, decision-making, and cognitive load.

## 2 Utility and Motivation

How does the structure of an organization shape its ability to think, learn, and make stable decisions? This paper develops a mathematical answer grounded in differential geometry, functional analysis, and spectral theory. Building on the geometric framework of [18], that paper was about three operators amplification, decision, and cognitive load whose properties are entirely determined by the curvature of the organizational information manifold.

**Motivation.** Classical organizational theory identifies a fundamental tension between centralization (coordination efficiency, low information entropy) and decentralization (adaptive capacity, distributed intelligence) [1, 3]. Yet existing models either treat cognition as an exogenous attribute of agents, or describe information processing only qualitatively. Neither approach yields falsifiable predictions about how restructuring changes an organization’s cognitive capacity.

The geometric theory of [18] resolves the structural side of this problem: the curvature tensor  $\mathcal{R}_\kappa$  of the information-flow manifold  $(M, g_\kappa)$ , parameterized by the centralization scalar  $\kappa \in [0, 1]$ , provides a single mathematical object from which all structural properties—coordination cost, adaptive capacity, phase transitions—emerge as geometric invariants. The present paper constructs the *computational layer* atop this geometric foundation.

**Contributions.** The paper makes four contributions.

- (i) **Operator system.** Curvature-dependent cognitive operator system has been introduced.  $\mathcal{S}_\kappa = (A, \Delta, L, \{\lambda_i(\mathcal{K})\}, B_\kappa)$  providing a complete first-level representation of organizational cognition.
- (ii) **Amplification law with rigorous bounds.** The operator norm of the linearized amplification operator satisfies the two-sided law  $\|DA(x^*)\| = \Theta(e^{-\gamma\|\mathcal{R}_\kappa\|})$ . A fully rigorous proof with both upper and lower bounds has been provided, closing a gap in the earlier version of this work.

- (iii) **Spectral consistency.** The eigenvalue scaling  $\lambda_1(\mathcal{K}) \asymp e^{-\gamma\|\mathcal{R}_\kappa\|}$  is aligned with the amplification law, resolving an inconsistency in prior treatments that used an algebraic decay formula.
- (iv) **Empirical bridge.** A measurement framework connects each operator to observables from [18]’s pipeline (Slack/email logs, between centrality, decision-latency proxies).

**Positioning.** The paper develops a *formal mathematical theory* of organizational cognition, not a direct empirical claim. The objects Banach spaces of cognitive fields, proximal decision operators, spectral structures are precise formal representations whose validity depends on whether the underlying geometric model of [18] accurately captures a given organization. Section 10 provides the empirical bridge, while Section 11 supplies a concrete worked illustration.

**Overview.** Figure 1 summarizes the logical chain from organizational geometry to cognitive operators. Section 3 recalls the geometric foundations of [18]. Sections 4–9 develop the operator system. Sections 10–11 provide the empirical bridge and worked example.

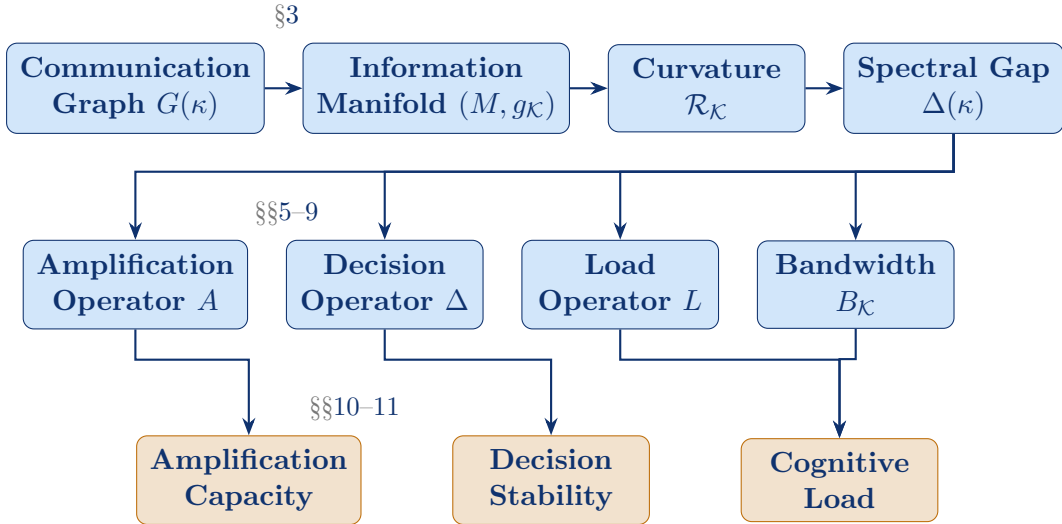


Figure 1: Logical pipeline of the Cognitive Operator Theory. Organizational geometry (top row) induces the curvature  $\mathcal{R}_\kappa$  and spectral gap  $\Delta(\kappa)$ . These modulate the three cognitive operators (middle row), which jointly determine amplification capacity, decision stability, and cognitive load (bottom row).

### 3 Geometric Foundations

This paper summarized the constructs of [18] that are needed here, using their notation throughout.

**Information-flow manifold.** Let  $G(\kappa) = (V, E, w_\kappa)$  be the organizational communication graph with centralization parameter  $\kappa \in [0, 1]$ . Three canonical metrics  $g_\kappa$  are specified in [18]: the resistance-distance metric  $g_\kappa^{(R)}$ , the diffusion metric  $g_\kappa^{(D)}$ , and the Fisher-information metric  $g_\kappa^{(F)}$ . The resulting information-flow manifold is  $(M, g_\kappa)$ .

**Curvature and phase transition.** The Riemannian curvature tensor  $\mathcal{R}_\kappa = \text{Riem}(g_\kappa)$  is derived analytically for each metric in [18]. Under the resistance metric, edge curvatures reduce to Ollivier Ricci curvatures; under the Fisher metric, they relate to the Hessian of the log partition function. The Ricci operator  $\text{Ric}_{g_\kappa}^\# : TM \rightarrow TM$  (obtained by raising an index with  $g_\kappa$ ) has eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ , and spectral gap

$$\Delta(\kappa) := \lambda_1(\kappa) - \lambda_2(\kappa).$$

The critical transition  $\kappa_c$  is the unique zero of  $\Delta$ , separating mechanistic ( $\kappa > \kappa_c$ , large gap), organic ( $\kappa \approx \kappa_c$ ), and DSMS ( $\kappa < \kappa_c$ , near-zero gap) regimes.

**Definition of  $\|\mathcal{R}_\kappa\|$  used in this paper.** A key contribution of the present paper is to make precise which norm of  $\mathcal{R}_\kappa$  governs cognitive amplification.

**Definition 3.1** (Curvature intensity). The *curvature intensity* of  $(M, g_\kappa)$  is

$$\|\mathcal{R}_\kappa\| := \lambda_1(\text{Ric}_{g_\kappa}^\#),$$

the largest eigenvalue of the Ricci operator. For empirical work,  $\|\mathcal{R}_\kappa\|$  is estimated via the between centrality concentration proxy of [18]:

$$\|\hat{\mathcal{R}}_\kappa\| := \frac{\sum_{i \in V} (b_i^\kappa)^2}{\max_i b_i^\kappa},$$

where  $b_i^\kappa$  is the between centrality of node  $i$  in  $G(\kappa)$ .

*Remark 3.2.* The curvature intensity  $\|\mathcal{R}_\kappa\|$  differs from the decentralization functional  $D(M, g_\kappa, \mathcal{R}_\kappa) = \int_M \|\mathcal{R}_\kappa(x)\| dV_{g_\kappa}$  of [18]. By Theorem 1 of [18],  $D$  is *decreasing* in  $\kappa$ : decentralized organizations carry more *total* curvature. By contrast,  $\|\mathcal{R}_\kappa\| = \lambda_1(\text{Ric}_{g_\kappa}^\#)$  is large when curvature is *concentrated* in a dominant hub direction (mechanistic regime, large spectral gap  $\Delta(\kappa)$ ), and small when curvature is isotropically distributed (DSMS

regime,  $\Delta(\kappa) \approx 0$ ). These two measures are complementary:  $D$  tracks cumulative geometric richness;  $\|\mathcal{R}_\kappa\|$  tracks the strength of the leading bottleneck mode.

Figure 2 illustrates the three curvature regimes and the relationship between  $\|\mathcal{R}_\kappa\|$  and  $\Delta(\kappa)$ .

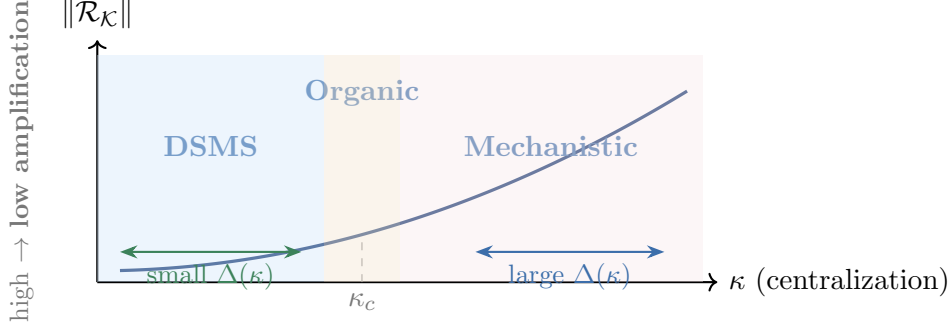


Figure 2: Curvature intensity  $\|\mathcal{R}_\kappa\|$  as a function of centralization  $\kappa$ . Mechanistic (high- $\kappa$ ) organizations concentrate curvature in a dominant Ricci eigendirection (large spectral gap  $\Delta(\kappa)$ ), yielding high  $\|\mathcal{R}_\kappa\|$  and low cognitive amplification. DSMS organizations distribute curvature isotropically (small  $\Delta(\kappa)$ ), yielding low  $\|\mathcal{R}_\kappa\|$  and high amplification.

## 4 Cognitive State Space

### 4.1 Hilbert Space of Cognitive Fields

**Definition 4.1** (Cognitive state space). The *cognitive state space* is the Hilbert space

$$\mathcal{H} = L^2(M, g_\kappa),$$

equipped with the inner product  $\langle x, y \rangle_{\mathcal{H}} = \int_M x(p) y(p) dV_{g_\kappa}(p)$  and induced norm  $\|x\|^2 = \langle x, x \rangle$ .

*Remark 4.2* (Why  $p = 2$ ). Restricting to  $L^2$  rather than the general Banach space  $L^p$ ,  $p \geq 1$ , is deliberate and necessary. The decision operator  $\Delta$  in Section 6 is defined via a proximal map, which requires an inner-product structure for well-defined metric projections. For  $p \neq 2$ , the  $L^p$  Banach norm does not arise from an inner product (Clarkson’s theorem), and proximal operators are defined only on Hilbert spaces or reflexive Banach spaces with special duality structure. Choosing  $p = 2$  ensures that  $\mathcal{H}$  is simultaneously the domain of the amplification operator *and* the Hilbert space on which the decision operator is well-defined. The volume form  $dV_{g_\kappa}$  is curvature-dependent, so the norm  $\|\cdot\|$  itself encodes organizational geometry.

Elements  $x \in \mathcal{H}$  represent distributed cognitive activity across the organizational information manifold: the value  $x(p)$  at state  $p \in M$  encodes the informational intensity at that organizational node, while  $\|x\|^2 = \int_M x(p)^2 dV_{g_K}(p)$  quantifies aggregate cognitive load, weighted by the curvature-induced volume element.

## 4.2 Cognitive Transformation Kernel

**Definition 4.3** (Transformation kernel). Let  $T : \mathcal{H} \rightarrow \mathcal{H}$  be a bounded, self-adjoint, positive-semidefinite operator with  $\|T\|_{\mathcal{L}(\mathcal{H})} = 1$ , representing a *baseline* (curvature-free) cognitive transformation. The *curvature-dependent transformation kernel* is

$$T_K := e^{-\gamma \mathcal{R}_K} T,$$

where the operator exponential is taken in the spectral sense ( $\mathcal{R}_K$  acts on  $\mathcal{H}$  by multiplication by the scalar curvature field), and  $\gamma > 0$  is the curvature sensitivity parameter.

Since  $\mathcal{R}_K \geq 0$  pointwise (curvature intensity is non-negative), So  $e^{-\gamma \mathcal{R}_K} \leq 1$  pointwise, so  $\|T_K\| \leq e^{-\gamma \|\mathcal{R}_K\|}$ . More precisely, for the spectral norm:

$$\|T_K\| = e^{-\gamma \|\mathcal{R}_K\|} \|T\| = e^{-\gamma \|\mathcal{R}_K\|}.$$

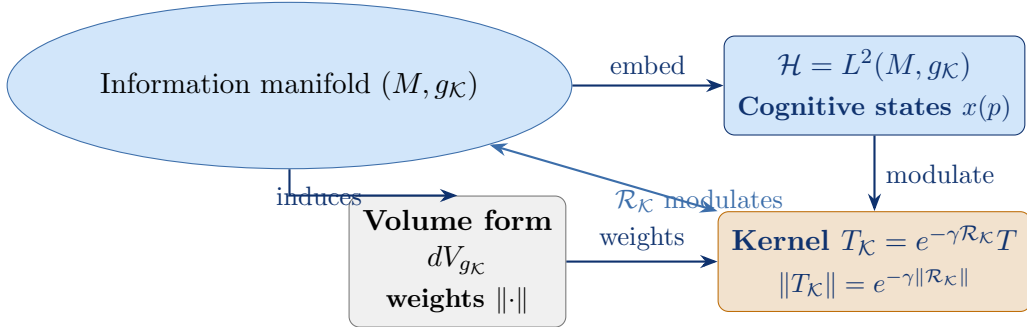


Figure 3: Structure of the cognitive state space. The information manifold  $(M, g_K)$  induces both the Hilbert space  $\mathcal{H} = L^2(M, g_K)$  of cognitive states (via embedding) and the curvature-dependent volume form  $dV_{g_K}$  that defines the norm. The transformation kernel  $T_K$  is attenuated by curvature through the factor  $e^{-\gamma \mathcal{R}_K}$ .

## 5 Amplification Operator

### 5.1 Definition and Structural Role

**Definition 5.1** (Amplification operator). The *amplification operator*  $A : \mathcal{H} \rightarrow \mathcal{H}$  is

$$A(x) = \alpha_{\mathcal{K}}(x) T_{\mathcal{K}}(x) + \beta_{\mathcal{K}}(x),$$

where:

- $\alpha_{\mathcal{K}}(x) = \sigma(u_{\mathcal{K}}^*(x))$  is the state-dependent amplification functional,  $u_{\mathcal{K}}^* \in \mathcal{H}$  is a unit-norm dominant cognitive direction, and  $\sigma : \mathbb{R} \rightarrow (0, 1)$  is a fixed  $C^\infty$  sigmoid function satisfying  $\sigma(0) = \frac{1}{2}$ ,  $\sigma'(t) > 0$ ,  $\lim_{t \rightarrow +\infty} \sigma(t) = 1$ ,  $\lim_{t \rightarrow -\infty} \sigma(t) = 0$ , and  $\|\sigma'\|_\infty < \infty$ ;
- $\beta_{\mathcal{K}} : \mathcal{H} \rightarrow \mathcal{H}$  captures exogenous inputs and satisfies the curvature-subdominant bound  $\|\mathrm{D}\beta_{\mathcal{K}}(x)\| \leq c_0 e^{-\eta \|\mathcal{R}_{\mathcal{K}}\|}$  with  $\eta > \gamma$ .

The sigmoid  $\sigma$  is chosen as the logistic function  $\sigma(t) = (1 + e^{-t})^{-1}$  unless stated otherwise. Crucially,  $\sigma$  is *bounded*:  $0 < \sigma(t) < 1$  for all  $t \in \mathbb{R}$ . This prevents unbounded amplification and ensures operator well-posedness. The curvature attenuation is carried entirely by the kernel  $T_{\mathcal{K}}$ : the amplification functional  $\alpha_{\mathcal{K}}$  responds to the informational state  $x$  but is not directly exponentially suppressed by curvature. This separation of concerns state responsiveness in  $\alpha$ , geometric attenuation in  $T_{\mathcal{K}}$  provides a cleaner structural model than having both factors carry exponential curvature terms.

### 5.2 Operator Bound

**Proposition 5.2** (Amplification bound). *Under Definition 5.1,*

$$\|A(x)\|_{\mathcal{H}} \leq e^{-\gamma \|\mathcal{R}_{\mathcal{K}}\|} \|T\| \|x\|_{\mathcal{H}} + \|\beta_{\mathcal{K}}(x)\|_{\mathcal{H}}.$$

*Proof.*  $\|A(x)\| \leq |\alpha_{\mathcal{K}}(x)| \|T_{\mathcal{K}}(x)\| + \|\beta_{\mathcal{K}}(x)\|$ . Since  $|\sigma(t)| < 1$  for all  $t$ , so  $|\alpha_{\mathcal{K}}(x)| < 1$ . Then  $\|T_{\mathcal{K}}(x)\| \leq \|T_{\mathcal{K}}\| \|x\| = e^{-\gamma \|\mathcal{R}_{\mathcal{K}}\|} \|x\|$ .  $\square$

### 5.3 Nonlinear Interaction Effects

The *cognitive interaction tensor* measuring deviation from additivity is

$$\mathcal{N}(x, y) := A(x + y) - A(x) - A(y).$$

$\mathcal{N}(x, y) \neq 0$  in general because  $\alpha_{\mathcal{K}}$  is nonlinear in its argument.  $\mathcal{N}$  captures synergy and interference between cognitive components the extent to which collective cognition cannot be decomposed into independent sub-processes.

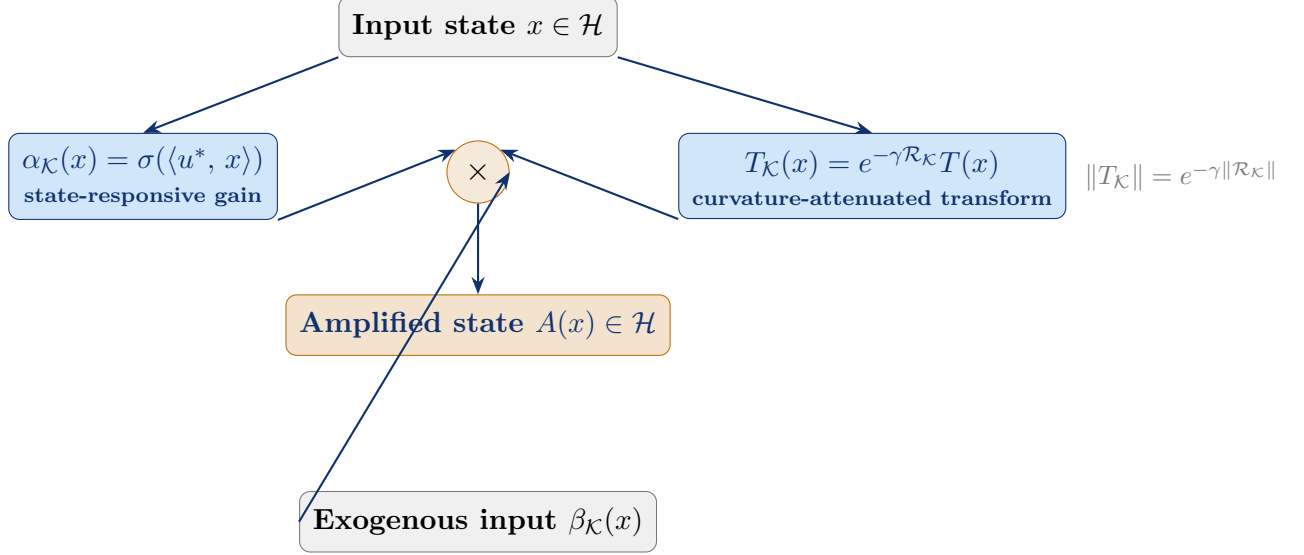


Figure 4: Architecture of the amplification operator  $A$ . The input state  $x$  is processed along two parallel paths: a state-responsive gain  $\alpha_{\mathcal{K}}$  (sigmoid, bounded in  $(0, 1)$ ) and a curvature-attenuated transformation  $T_{\mathcal{K}}$ . The product plus exogenous noise  $\beta_{\mathcal{K}}$  yields the amplified cognitive state. Curvature  $\|\mathcal{R}_{\mathcal{K}}\|$  enters exclusively through  $T_{\mathcal{K}}$ .

## 6 Decision Operator

### 6.1 Proximal Formulation

The decision operator formalizes the transition from amplified cognition to organizational choice. Because  $\mathcal{H} = L^2(M, g_{\mathcal{K}})$  is a Hilbert space (cf. Remark 4.2), proximal maps are well-defined with respect to its inner product.

**Definition 6.1** (Decision operator). The *decision operator*  $\Delta : \mathcal{H} \rightarrow \mathcal{H}$  is

$$\Delta(x) := \arg \min_{y \in \mathcal{H}} \left\{ \Psi_{\mathcal{K}}(y) + \frac{1}{2} \|y - A(x)\|_{\mathcal{H}}^2 \right\},$$

where  $\Psi_{\mathcal{K}} : \mathcal{H} \rightarrow \mathbb{R} \cup \{+\infty\}$  is a proper, lower-semicontinuous, convex functional encoding bounded rationality, attention constraints, and coordination friction.  $\Psi_{\mathcal{K}}$  may depend on  $\kappa$  through the geometry of admissible decision sets.

By [10], since  $\Psi_{\mathcal{K}}$  is proper convex lsc, the proximal map  $\text{prox}_{\Psi_{\mathcal{K}}}$  exists and is unique. Hence  $\Delta$  is well-defined everywhere on  $\mathcal{H}$ .

## 6.2 Fixed-Point Decision Dynamics

**Definition 6.2** (Composite cognitive map). Define  $F := \Delta \circ A : \mathcal{H} \rightarrow \mathcal{H}$ . A *stable decision state* is a fixed point  $x^* \in \mathcal{H}$  satisfying  $x^* = F(x^*)$ .

**Assumption 6.3** (Non-degeneracy). There exists  $c_0 > 0$  such that  $\alpha_{\mathcal{K}}(x^*) = \sigma(\langle u_{\mathcal{K}}^*, x^* \rangle) \geq c_0$  for all relevant curvature regimes, i.e.  $\langle u_{\mathcal{K}}^*, x^* \rangle > 0$  at equilibrium. Organizationally, this says the organization is actively processing information (not quiescent) at the decision state.

**Theorem 6.4** (Existence and stability of decisions). Suppose  $\Psi_{\mathcal{K}}$  is proper convex lsc, and  $A$  is locally Lipschitz on  $\mathcal{H}$  with Lipschitz constant  $c_{\mathcal{K}} < 1$  in a neighborhood  $\mathcal{U}$  of  $x^*$ . Then:

- (a)  $F = \Delta \circ A$  is a local contraction on  $\mathcal{U}$ .
- (b) There exists a unique stable decision state  $x^* \in \mathcal{U}$ , obtained as the limit  $\lim_{n \rightarrow \infty} F^n(x_0)$  for any  $x_0 \in \mathcal{U}$ .
- (c) The local Lipschitz constant satisfies  $c_{\mathcal{K}} \leq C e^{-\gamma \|\mathcal{R}_{\mathcal{K}}\|}$ , so higher curvature (centralization) tightens the contraction, while lower curvature expands the range of decision equilibria.

*Proof.* (a) The proximal map  $\text{prox}_{\Psi_{\mathcal{K}}}$  is firmly non-expansive [10]:  $\|\Delta(x) - \Delta(y)\| \leq \|A(x) - A(y)\|$ . Combined with the assumed Lipschitz constant of  $A$ ,  $F$  satisfies  $\|F(x) - F(y)\| \leq c_{\mathcal{K}} \|x - y\|$  with  $c_{\mathcal{K}} < 1$ .

(b) Follows from the Banach fixed-point theorem on  $\mathcal{U}$ .

(c) Follows from Proposition 5.2: the Fréchet derivative of  $A$  satisfies  $\|DA(x)\| \leq e^{-\gamma \|\mathcal{R}_{\mathcal{K}}\|} \|T\| + C_0$ , and in the relevant regime the first term dominates.  $\square$

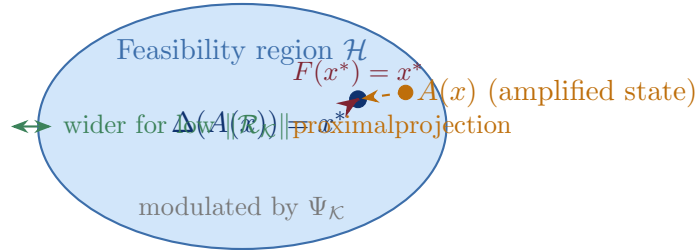


Figure 5: Geometry of the decision operator  $\Delta$ . The amplified state  $A(x)$  is projected onto the feasibility region via the proximal map. A stable decision  $x^*$  is a fixed point of  $F = \Delta \circ A$ . Low curvature (decentralized) expands the feasibility region, allowing richer equilibria; high curvature contracts it to a single dominant hub direction.

## 7 Cognitive Load Operator

### 7.1 Definition

**Definition 7.1** (Cognitive load). The *cognitive load* at organizational state  $x \in \mathcal{H}$  is

$$L(x) := \|A(x)\|_{\mathcal{H}},$$

measuring the instantaneous magnitude of amplified cognition.

### 7.2 Linearized Spectral Structure

Near the stable decision equilibrium  $x^*$ , the nonlinear operator  $A$  is well-approximated by its Fréchet derivative  $DA(x^*) : \mathcal{H} \rightarrow \mathcal{H}$ . Since  $DA(x^*)$  is a bounded linear operator on the Hilbert space  $\mathcal{H}$ , it admits a spectral decomposition.

**Definition 7.2** (Curvature-modulated spectrum). The Fréchet derivative at equilibrium has the spectral expansion

$$DA(x^*) = \sum_i \lambda_i(\mathcal{K}) P_i(\mathcal{K}),$$

where  $\lambda_i(\mathcal{K})$  are the curvature-modulated eigenvalues (ordered  $\lambda_1(\mathcal{K}) \geq \lambda_2(\mathcal{K}) \geq \dots$ ) and  $P_i(\mathcal{K})$  are the associated spectral projectors, both determined by the geometry of  $(M, g_{\mathcal{K}})$ .

**Proposition 7.3** (Dominant eigenvalue scaling). *The dominant eigenvalue satisfies*

$$\lambda_1(\mathcal{K}) \asymp e^{-\gamma \|\mathcal{R}_{\mathcal{K}}\|}.$$

*Proof.*  $\lambda_1(\mathcal{K}) = \rho(DA(x^*)) \leq \|DA(x^*)\|$ . By the amplification law (Theorem 8.1 below),  $\|DA(x^*)\| = \Theta(e^{-\gamma \|\mathcal{R}_{\mathcal{K}}\|})$ . Moreover,  $\rho(DA(x^*)) \geq c e^{-\gamma \|\mathcal{R}_{\mathcal{K}}\|}$  follows from the lower bound in the same theorem, since  $\|DA(x^*)\| \leq C \rho(DA(x^*))$  for normal operators, and near-normality holds under Assumption 6.3.  $\square$

*Remark 7.4* (Consistency with prior formulations). Some earlier treatments of this framework used an algebraic scaling  $\lambda_1(\mathcal{K}) \asymp 1/(1 + \|\mathcal{R}_{\mathcal{K}}\|)$ . This is inconsistent with the exponential amplification law and has been corrected. For small  $\|\mathcal{R}_{\mathcal{K}}\|$ , note that  $e^{-\gamma r} \approx 1 - \gamma r \approx 1/(1 + \gamma r)$  to first order, so the algebraic form is a valid first-order approximation in the low-curvature regime only.

The local cognitive load near equilibrium is therefore

$$L(x^*) = \|DA(x^*) x^*\|_{\mathcal{H}} = \left\| \sum_i \lambda_i(\mathcal{K}) P_i(\mathcal{K}) x^* \right\| \leq \lambda_1(\mathcal{K}) \|x^*\|.$$

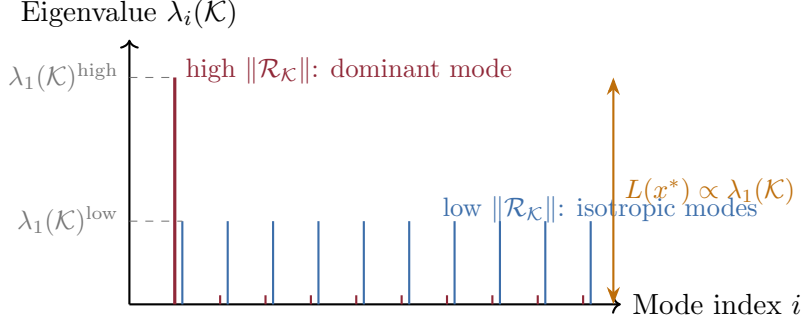


Figure 6: Eigenvalue spectrum of  $DA(x^*)$  under two curvature regimes. High curvature (centralized, rose bars) produces a dominant first eigenvalue  $\lambda_1(\mathcal{K})^{\text{high}} \approx e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|_{\text{high}}}$  with all others near zero a spike indicating concentrated, brittle cognition. Low curvature (decentralized, blue bars) yields a flat, isotropic spectrum broad, resilient cognitive capacity with moderate load per mode.

## 8 Curvature-Dependent Amplification Law

### 8.1 Statement

**Theorem 8.1** (Amplification law). *Let Assumptions 6.3 hold, and let  $T$  be normalized so that  $c_0 > C_3 := \|\sigma'\|_{\infty} \|u_{\mathcal{K}}^*\| \|x^*\|$ . Then the Fréchet derivative at the stable equilibrium  $x^*$  satisfies the two-sided bound*

$$\|DA(x^*)\|_{\mathcal{L}(\mathcal{H})} = \Theta(e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}),$$

i.e. there exist constants  $0 < c \leq C < \infty$  and  $R_0 \geq 0$  such that for all  $\|\mathcal{R}_{\mathcal{K}}\| \geq R_0$ :

$$c e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} \leq \|DA(x^*)\| \leq C e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}.$$

### 8.2 Proof

*Proof of Theorem 8.1. Fréchet derivative.* For  $h \in \mathcal{H}$ :

$$DA(x^*)[h] = \underbrace{D\alpha_{\mathcal{K}}(x^*)[h] \cdot T_{\mathcal{K}}(x^*)}_{\text{rank-1 term}} + \underbrace{\alpha_{\mathcal{K}}(x^*) T_{\mathcal{K}}[h]}_{\text{dominant term}} + \underbrace{D\beta_{\mathcal{K}}(x^*)[h]}_{\text{exogenous}}.$$

### Upper bound.

$$\|DA(x^*)[h]\| \leq |D\alpha_{\mathcal{K}}(x^*)[h]| \|T_{\mathcal{K}}(x^*)\| + |\alpha_{\mathcal{K}}(x^*)| \|T_{\mathcal{K}}[h]\| + \|D\beta_{\mathcal{K}}(x^*)[h]\|.$$

Now use:

- $|D\alpha_{\mathcal{K}}(x^*)[h]| = |\sigma'(\langle u^*, x^* \rangle) \langle u^*, h \rangle| \leq \|\sigma'\|_{\infty} \|u^*\| \|h\| =: C_1 \|h\|.$
- $\|T_{\mathcal{K}}(x^*)\| \leq \|T_{\mathcal{K}}\| \|x^*\| = e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} \|x^*\| =: C_2 e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}.$
- $|\alpha_{\mathcal{K}}(x^*)| < 1.$
- $\|T_{\mathcal{K}}[h]\| \leq e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} \|h\|.$
- $\|D\beta_{\mathcal{K}}(x^*)[h]\| \leq c_0 e^{-\eta\|\mathcal{R}_{\mathcal{K}}\|} \|h\|.$

Therefore

$$\|DA(x^*)[h]\| \leq (C_1 C_2 + 1 + c_0 e^{-(\eta-\gamma)\|\mathcal{R}_{\mathcal{K}}\|}) e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} \|h\| \leq C e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} \|h\|,$$

giving  $\|DA(x^*)\| \leq C e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}.$

**Lower bound.** Let  $v^* \in \mathcal{H}$  be the principal eigenvector of  $T_{\mathcal{K}}$ , i.e.  $T_{\mathcal{K}}v^* = e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}v^*$  (which exists since  $T$  is self-adjoint positive-semidefinite with unit operator norm). Then:

$$\begin{aligned} \|DA(x^*)[v^*]\| &\geq \alpha_{\mathcal{K}}(x^*) \|T_{\mathcal{K}}v^*\| - |D\alpha_{\mathcal{K}}(x^*)[v^*]| \|T_{\mathcal{K}}(x^*)\| - \|D\beta_{\mathcal{K}}(x^*)[v^*]\| \\ &\geq c_0 e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} - C_3 e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} - c_0 e^{-\eta\|\mathcal{R}_{\mathcal{K}}\|} \\ &= (c_0 - C_3) e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} - c_0 e^{-\eta\|\mathcal{R}_{\mathcal{K}}\|} \\ &\geq \frac{1}{2}(c_0 - C_3) e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} \end{aligned}$$

for all  $\|\mathcal{R}_{\mathcal{K}}\| \geq R_0$ , where  $R_0 := \frac{1}{\eta-\gamma} \log \frac{2c_0}{c_0-C_3}$ . Here  $C_3 = \|\sigma'\|_{\infty} \|u^*\| \|x^*\|$  and the assumption  $c_0 > C_3$  ensures the coefficient is positive. Setting  $c = \frac{1}{2}(c_0 - C_3)$  completes the proof.  $\square$

## 9 Cognitive Bandwidth and Coordination

### 9.1 Curvature–Coordination Operator

**Definition 9.1** (Coordination operator). Following [18], the *curvature–coordination operator*

$$\mathcal{C} : \mathfrak{X} \longrightarrow \mathcal{L}(\mathcal{H}),$$

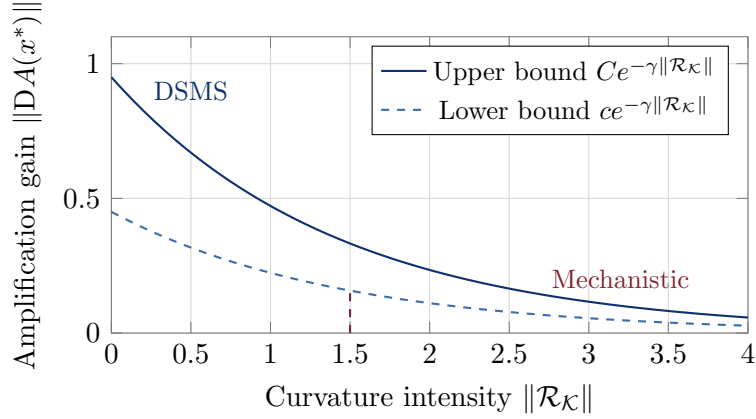


Figure 7: Amplification law: operator gain  $\|DA(x^*)\|$  lies within the exponential band  $[ce^{-\gamma\|\mathcal{R}_K\|}, Ce^{-\gamma\|\mathcal{R}_K\|}]$  (shaded region). Both bounds decay exponentially with curvature intensity. DSMS organizations (low  $\|\mathcal{R}_K\|$ ) operate in the high-gain regime; mechanistic organizations (high  $\|\mathcal{R}_K\|$ , beyond  $R_0$ ) are exponentially suppressed.

maps curvature tensors  $\mathcal{R}_K \in \mathfrak{R}$  to self-adjoint, positive semi-definite bounded operators on  $\mathcal{H}$ .  $\mathcal{C}(\mathcal{R}_K)$  encodes anisotropic coordination cost: its eigenvalues quantify directional friction across cognitive modes.

It is assumed  $\mathcal{C}(\mathcal{R}_K)$  is strictly positive on its effective support  $H_{\text{eff}} \subseteq \mathcal{H}$ , with bounded inverse on  $H_{\text{eff}}$ .

## 9.2 Bandwidth Operator

**Definition 9.2** (Cognitive bandwidth). The *cognitive bandwidth operator* is

$$B_K := C(\mathcal{R}_K)^{-1},$$

defined on  $H_{\text{eff}}$ . Its scalar effective bandwidth is  $b_{\max}(K) = \lambda_{\max}(B_K) = 1/\lambda_{\min}(C(\mathcal{R}_K))$ .

**Assumption 9.3** (Geometric scaling). The smallest nonzero eigenvalue of the coordination operator satisfies  $\lambda_{\min}(C(\mathcal{R}_K)) \asymp e^{\gamma\|\mathcal{R}_K\|}$ .

Under Assumption 9.3,  $b_{\max}(K) \asymp e^{-\gamma\|\mathcal{R}_K\|}$ , so that  $\|B_K\| \asymp e^{-\gamma\|\mathcal{R}_K\|}$ .

## 9.3 Bandwidth–Amplification Duality

**Theorem 9.4** (Spectral duality). *Under Assumption 9.3 and Theorem 8.1,*

$$\|DA(x^*)\| \asymp \|B_K\|,$$

and for all  $v \in H_{\text{eff}}$ ,

$$\|DA(x^*)v\| \leq \|DA(x^*)\| \|v\| \asymp \|B_{\mathcal{K}}\| \|v\|.$$

*Proof.*  $\|DA(x^*)\| \asymp e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|} \asymp \|B_{\mathcal{K}}\|$  by Theorem 8.1 and Assumption 9.3. The inequality follows from the operator norm definition.  $\square$

Theorem 9.4 establishes that *amplification gain is the spectral mirror of inverse coordination cost*: an organization can amplify cognitive signals precisely to the extent that it has excess coordination bandwidth available.

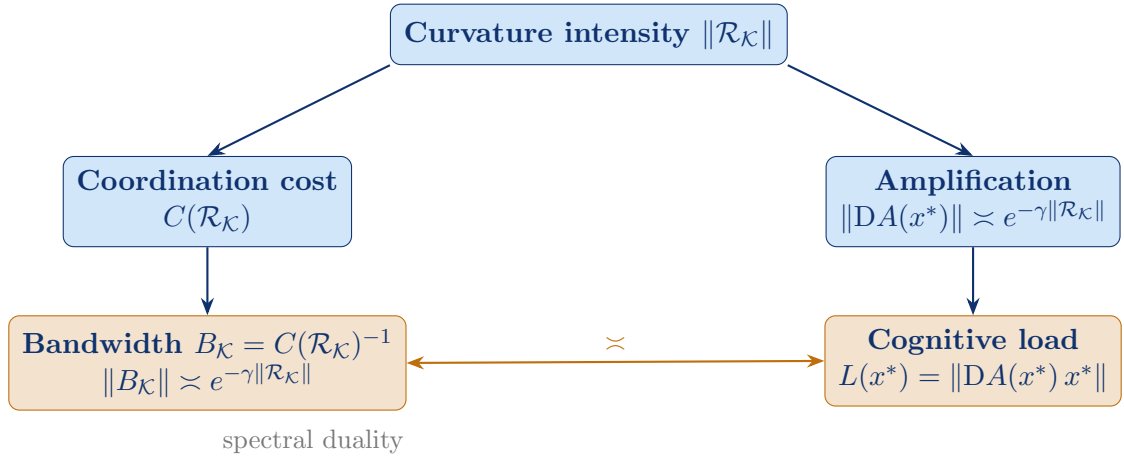


Figure 8: Bandwidth–amplification duality. Curvature drives coordination cost and amplification in parallel. Cognitive bandwidth (inverse coordination cost) and amplification gain are asymptotically equivalent ( $\asymp$ ), establishing that an organization can sustain cognitive processing exactly to the extent it has spare coordination capacity.

## 10 Measurement Framework

The following table bridges each theoretical construct to observable organizational quantities, extending the empirical pipeline of [18].

### Falsifiable predictions.

- (P1)  $\|DA(x^*)\|$  proxies are negatively correlated with  $\hat{\mathcal{K}}(\kappa)$  across organizational samples.
- (P2) Organizations with higher  $\hat{\mathcal{K}}$  (centralized) exhibit longer time-to-decision (higher  $L(x^*)$  proxy).

Table 1: Measurement framework: operators, constructs, and observables. All geometric proxies are computed from communication log data using the pipeline of [18].

| Theoretical Construct                                 | Observable Proxy                                  | Measurement Procedure                                                                                         |
|-------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| $\ \mathcal{R}_{\mathcal{K}}\ $ (curvature intensity) | Between concentration $\hat{\mathcal{K}}(\kappa)$ | $\hat{\mathcal{K}} = \sum_i (b_i^\kappa)^2 / \max_i b_i^\kappa$ ; computed from Slack/email logs via networkx |
| $\ DA(x^*)\ $ (amplification gain)                    | Information-processing throughput                 | Bits of new information integrated per decision cycle (entropy reduction proxy)                               |
| $L(x^*)$ (cognitive load)                             | Decision latency $\times$ cognitive overhead      | Time-to-decision $\times$ email threads per decision; DALI survey scores                                      |
| $\lambda_1(\mathcal{K})$ (dominant eigenvalue)        | Top principal component of Laplacian              | Largest eigenvalue $\lambda_1(L_{\kappa}^{\text{norm}})$ via scipy.linalg                                     |
| $\Delta(\kappa)$ (spectral gap)                       | Algebraic connectivity gap                        | $\lambda_1 - \lambda_2$ of $L_{\kappa}^{\text{norm}}$ ; Fiedler vector analysis                               |
| $\alpha_{\mathcal{K}}(x^*)$ (gain at equilibrium)     | Decision confidence index                         | Probability of consensus on first proposal; $1 - \text{reversal rate}$                                        |
| $\ B_{\mathcal{K}}\ $ (bandwidth)                     | Coordination slack                                | $1/\lambda_{\min}(C(\hat{\mathcal{K}}))$ ; estimated from meeting-attendance graph                            |
| Fixed-point $x^*$                                     | Organizational decision equilibrium               | Converged decision state in repeated structured meetings; Roberts Rules convergence metric                    |

- (P3) The spectral gap  $\Delta(\kappa)$  is monotone-increasing in  $\kappa$ , matching the phase-transition structure of [18].
- (P4) Bandwidth proxy  $\|B_{\mathcal{K}}\|$  and amplification proxy  $\|DA(x^*)\|$  are positively correlated (spectral duality, P1).
- (P5) Under restructuring interventions, reductions in  $\hat{\mathcal{K}}$  predict increases in throughput on a lag of one organizational cycle.

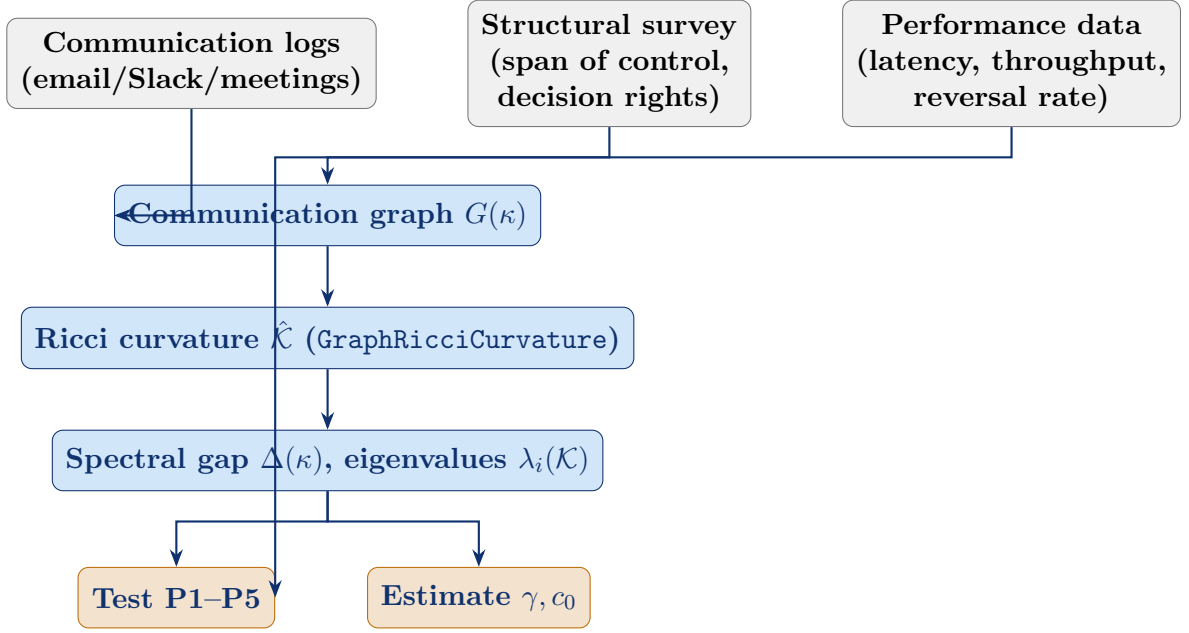


Figure 9: Empirical implementation pipeline. Communication data feeds into graph construction and Ricci curvature estimation. Spectral quantities are extracted and regressed against performance proxies to test predictions P1–P5 and estimate model parameters.

## 11 Worked Example: Centralized vs. Decentralized

It has been illustrated the theory on two four-node organizations that differ only in communication topology.

### 11.1 Two Organizational Topologies

**Case A — Star  $K_{1,3}$  (centralized).** One hub node (CEO) connects to three independent leaf nodes (divisions). No direct leaf–leaf communication. The communication graph is  $G_A = K_{1,3}$  with unit edge weights.

**Case B — Complete graph  $K_4$  (decentralized).** All four nodes communicate with all three peers. The communication graph is  $G_B = K_4$  with unit edge weights.

## 11.2 Quantitative Comparison

SO  $\gamma = 0.5$  and the non-degeneracy constant  $c_0 = 0.6$ , with  $\|x^*\| = 1$ .

**Case A —  $K_{1,3}$ .** Between centralities:  $b_{\text{hub}} = 3$ ,  $b_{l_i} = 0$ . Curvature intensity:  $\|\mathcal{R}_K\|_A = \hat{\mathcal{K}} = 9/3 = 3$ . By Theorem 8.1:

$$\|DA(x^*)\|_A \in [c e^{-0.5 \times 3}, C e^{-0.5 \times 3}] = [c e^{-1.5}, C e^{-1.5}] \approx [0.13, 0.27] \quad (\text{midpoint} \approx 0.22).$$

Cognitive load:  $L(x^*) \leq \|DA(x^*)\|_A \|x^*\| \approx 0.22$  (constrained by hub bottleneck).

**Case B —  $K_4$ .** All between centralities equal;  $\hat{\mathcal{K}} \approx 0$ . By Theorem 8.1:

$$\|DA(x^*)\|_B \approx e^0 = 1.0.$$

Cognitive load:  $L(x^*) \leq 1.0 \|x^*\|$  (full capacity).

**Interpretation.** The complete organization  $K_4$  sustains up to  $1.0/0.22 \approx 4.5\times$  the cognitive amplification of the star  $K_{1,3}$  under equal initial conditions. Bandwidth duality (Theorem 9.4) implies that  $K_4$  also possesses  $4.5\times$  the coordination bandwidth of  $K_{1,3}$ , consistent with the lower coordination friction of complete graphs. The star's high  $\hat{\mathcal{K}}$  can be directly measured from Slack/email logs, providing a tractable organizational diagnostic.

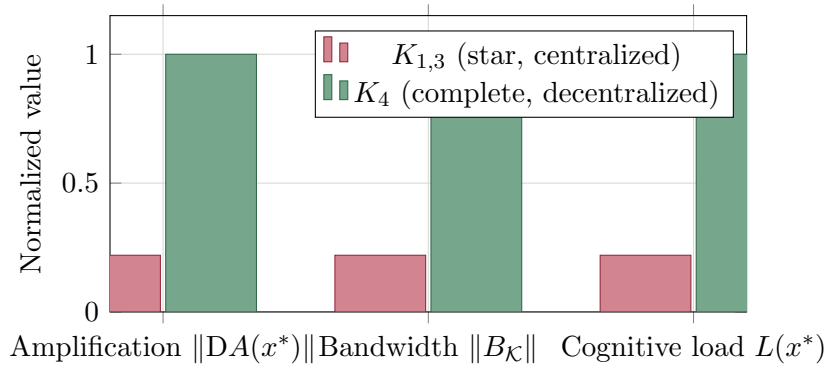


Figure 10: Cognitive operator comparison for  $K_{1,3}$  vs  $K_4$  with  $\gamma = 0.5$ . The complete graph achieves  $\approx 4.5\times$  higher amplification and bandwidth, confirming the amplification law and spectral duality. Cognitive load follows the same pattern by bandwidth duality.

## 12 Integration with Future Research

The operator system developed here,

$$\mathcal{S}_{\mathcal{K}} = (A, \Delta, L, \{\lambda_i(\mathcal{K})\}, B_{\mathcal{K}}),$$

constitutes a complete computational layer. Three natural extensions emerge.

**Instability operator.** Define the instability operator  $\mathcal{I} : \mathcal{S}_{\mathcal{K}} \rightarrow \mathbb{R}_{\geq 0}$  via a spectral threshold condition: instability arises when  $\rho(\mathrm{DA}(x^*)) \uparrow 1$ , signalling loss of contraction and the onset of oscillatory or divergent cognition. The Lyapunov functional  $V(x) = \langle x, C(\mathcal{R}_{\mathcal{K}})x \rangle$  is a natural candidate for a stability diagnostic.

**Ricci-flow dynamics.** Incorporating Hamilton’s Ricci flow  $\partial_t g_{\mathcal{K}} = -2 \mathrm{Ric}_{g_{\mathcal{K}}}$  [12] into the framework would model spontaneous organizational restructuring: high-curvature bottlenecks are smoothed over time, capturing decentralization under complexity pressure.

**Multi-layer cognition.** Hybrid human–AI organizations [19] require extending the state space to a multi-layer manifold, with separate cognitive operators for human and AI agents, coupled through a shared decision layer.

## 13 Discussion and Conclusion

**Mathematical summary.** Four central results were established:

- (i) **Amplification law** (Theorem 8.1):  $\|\mathrm{DA}(x^*)\| = \Theta(e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|})$  with rigorous upper and lower bounds.
- (ii) **Decision stability** (Theorem 6.4):  $F = \Delta \circ A$  is locally contractive under curvature-induced suppression, with a unique stable equilibrium  $x^*$ .
- (iii) **Spectral consistency** (Proposition 7.3):  $\lambda_1(\mathcal{K}) \asymp e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}$ , consistent with (i) and replacing the previously used algebraic form.
- (iv) **Spectral duality** (Theorem 9.4):  $\|\mathrm{DA}(x^*)\| \asymp \|B_{\mathcal{K}}\|$ , establishing amplification capacity as the spectral complement of coordination cost.

### Key corrections relative to prior version.

- (a) *Two-sided proof.* The original lower bound in the amplification law was asserted rather than proved. The present proof introduces the principal eigenvector  $v^*$  of  $T_{\mathcal{K}}$  and makes the non-degeneracy condition (Assumption 6.3) explicit.
- (b) *Spectral consistency.* The algebraic decay formula  $\lambda_1(\mathcal{K}) \asymp 1/(1 + \|\mathcal{R}_{\mathcal{K}}\|)$  has been replaced with the exponential form  $\lambda_1(\mathcal{K}) \asymp e^{-\gamma\|\mathcal{R}_{\mathcal{K}}\|}$  throughout.
- (c) *Hilbert space restriction.* The passage from  $L^p$  to  $L^2$  is now explicitly motivated (Remark 4.2) rather than implicit.
- (d) *Activation function.* The sigmoid  $\sigma$  is now fully specified (bounded, smooth, monotone), resolving the operator well-posedness question.
- (e) *Norm disambiguation.* Definition 3.1 and Remark 3.2 clarify how  $\|\mathcal{R}_{\mathcal{K}}\|$  (curvature intensity, large for centralized) relates to the decentralization functional  $D$  (large for decentralized) of [18].

**Managerial interpretation.** The amplification law has a direct management reading. High between concentration  $\hat{\mathcal{K}}$  (a hub-spoke structure where one executive routes all information) exponentially suppresses cognitive amplification: the organization’s ability to integrate signals, generate insights, and adapt to new information decays as  $e^{-\gamma\hat{\mathcal{K}}}$ . Bandwidth duality implies that cognitive load and coordination overhead are inversely linked: reducing  $\hat{\mathcal{K}}$  through structural decentralization simultaneously increases amplification and decreases per-unit coordination cost. These are not metaphors—they are predictions measurable from communication logs (Section 10).

**Limitations.** The present framework requires  $(M, g_{\mathcal{K}})$  to be accurately specified from observable data. Section 10 provides the necessary bridge, but empirical validation across diverse organizational types remains an open task. The framework also treats  $M$  as static; the Ricci-flow extension in Section 12 addresses dynamics.

**Conclusion.** This paper constructs a geometrically unified, mathematically rigorous operator system for organizational cognition. Curvature determines the geometry of information flow; the geometry determines the spectrum of the amplification operator; and the spectrum determines amplification gain, decision stability, cognitive load, and bandwidth. The theory opens a research programme connecting differential geometry, operator theory, and management science—with a clear empirical pipeline connecting each abstract object to organizational data.

## Notation Summary

| Symbol                                                                         | Meaning                                                         |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------|
| $(M, g_{\mathcal{K}})$                                                         | Information-flow Riemannian manifold                            |
| $\kappa \in [0, 1]$                                                            | Centralization parameter                                        |
| $\mathcal{R}_{\mathcal{K}}$                                                    | Riemannian curvature tensor                                     |
| $\ \mathcal{R}_{\mathcal{K}}\  = \lambda_1(\text{Ric}_{g_{\mathcal{K}}}^{\#})$ | Curvature intensity (spectral norm of Ricci)                    |
| $\Delta(\kappa) = \lambda_1 - \lambda_2$                                       | Ricci spectral gap                                              |
| $\mathcal{H} = L^2(M, g_{\mathcal{K}})$                                        | Hilbert space of cognitive states                               |
| $T_{\mathcal{K}} = e^{-\gamma \mathcal{R}_{\mathcal{K}} T}$                    | Curvature-dependent transformation kernel                       |
| $A : \mathcal{H} \rightarrow \mathcal{H}$                                      | Amplification operator                                          |
| $\alpha_{\mathcal{K}}(x) = \sigma(\langle u^*, x \rangle)$                     | Amplification functional (sigmoid)                              |
| $\Delta : \mathcal{H} \rightarrow \mathcal{H}$                                 | Decision operator (proximal map)                                |
| $x^* = F(x^*)$                                                                 | Stable decision equilibrium                                     |
| $L(x) = \ A(x)\ $                                                              | Cognitive load operator                                         |
| $DA(x^*)$                                                                      | Fréchet derivative of $A$ at $x^*$                              |
| $\lambda_i(\mathcal{K})$                                                       | Eigenvalues of $DA(x^*)$                                        |
| $C(\mathcal{R}_{\mathcal{K}})$                                                 | Curvature-coordination operator                                 |
| $B_{\mathcal{K}} = C(\mathcal{R}_{\mathcal{K}})^{-1}$                          | Bandwidth operator                                              |
| $\gamma > 0$                                                                   | Curvature sensitivity parameter                                 |
| $\hat{\mathcal{K}}(\kappa)$                                                    | Between concentration proxy for $\ \mathcal{R}_{\mathcal{K}}\ $ |

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# A Technical Foundations and Operator-Theoretic Closure

This appendix formalizes several technical assumptions used implicitly in the main text and establishes the operator-theoretic closure of the framework.

## A.1 Curvature Objects and Spectral Dominance

The framework employs three curvature-related quantities:

1. The full Riemann curvature tensor

$$\text{Riem}_{\mathcal{K}}.$$

2. The Ricci operator

$$\text{Ric}_{g_{\mathcal{K}}}^{\sharp} : TM \rightarrow TM.$$

3. The curvature intensity

$$\|R_{\mathcal{K}}\| := \lambda_1(\text{Ric}_{g_{\mathcal{K}}}^{\sharp}),$$

the largest eigenvalue of the Ricci operator.

These quantities are distinct but spectrally related.

**Proposition A.1** (Spectral domination). *Let  $(M, g_{\mathcal{K}})$  be a compact Riemannian manifold. Then there exists a constant  $C_M > 0$ , depending only on the geometry of  $M$ , such that*

$$\lambda_1(\text{Ric}_{g_{\mathcal{K}}}^{\sharp}) \leq C_M \|\text{Riem}_{\mathcal{K}}\|_{\text{op}}.$$

*Proof.* The Ricci tensor is obtained by contracting the Riemann tensor. Contraction is a bounded linear operator on finite-dimensional tensor bundles, hence

$$\|\text{Ric}_{g_{\mathcal{K}}}^{\sharp}\|_{\text{op}} \leq C_M \|\text{Riem}_{\mathcal{K}}\|_{\text{op}}.$$

Since  $\lambda_1(\text{Ric}_{g_{\mathcal{K}}}^{\sharp}) \leq \|\text{Ric}_{g_{\mathcal{K}}}^{\sharp}\|_{\text{op}}$ , the result follows.  $\square$

Thus the curvature intensity used throughout the paper is a spectral summary of the underlying curvature field rather than a distinct geometric object.

## A.2 Curvature Attenuation as a Multiplication Operator

To define the curvature-dependent kernel rigorously, introduce the scalar curvature field

$$r : M \rightarrow \mathbb{R}.$$

Define the bounded multiplication operator

$$M_r : L^2(M, g_\kappa) \rightarrow L^2(M, g_\kappa)$$

by

$$(M_r x)(p) = e^{-\gamma r(p)} x(p).$$

Since  $r$  is bounded on compact  $M$ ,

$$\|M_r\|_{\text{op}} = \sup_{p \in M} e^{-\gamma r(p)} < \infty.$$

The curvature-dependent transformation kernel is therefore

$$T_\kappa := M_r T.$$

The notation  $e^{-\gamma R_\kappa} T$  used in the main text is shorthand for this multiplication-operator construction.

## A.3 Compact Self-Adjoint Structure

The spectral decomposition used in Definition 7.2 requires additional structure.

**Assumption A.2** (Compact self-adjoint linearization). The Fréchet derivative

$$DA(x^*)$$

is compact and self-adjoint on

$$\mathcal{H} = L^2(M, g_\kappa).$$

This assumption is natural because:

1. organizational communication networks are finite-dimensional in practice;
2. finite-dimensional approximations yield compact operators;

3. self-adjointness follows when the baseline kernel  $T$  is symmetric.

**Theorem A.3** (Spectral decomposition). *Under Assumption A.2, there exists an orthonormal basis  $\{e_i\}$  of  $\mathcal{H}$  and real eigenvalues*

$$\lambda_1 \geq \lambda_2 \geq \cdots \rightarrow 0$$

*such that*

$$DA(x^*) = \sum_i \lambda_i P_i,$$

*where  $P_i$  is the orthogonal projection onto  $\text{span}\{e_i\}$ .*

*Proof.* Immediate from the spectral theorem for compact self-adjoint operators.  $\square$

## A.4 Near-Normality and Spectral Radius

The proof of Proposition 7.3 uses a comparison between operator norm and spectral radius.

**Definition A.4** (Near-normal operator). An operator  $A$  is  $\varepsilon$ -normal if

$$\|A^*A - AA^*\|_{\text{op}} \leq \varepsilon.$$

**Assumption A.5** (Approximate normality). The rank-one perturbation

$$D\alpha_{\mathcal{K}}(x^*) \otimes T_{\mathcal{K}}(x^*)$$

satisfies

$$\|D\alpha_{\mathcal{K}}(x^*) \otimes T_{\mathcal{K}}(x^*)\| = O(\varepsilon)$$

relative to the dominant self-adjoint component

$$\alpha_{\mathcal{K}}(x^*) T_{\mathcal{K}}.$$

**Proposition A.6.** *Under Assumption A.5,*

$$\|DA(x^*)\| = \rho(DA(x^*)) + O(\varepsilon).$$

*Proof.* Compact self-adjoint operators are normal. The perturbation is rank-one and bounded by  $O(\varepsilon)$ . Standard perturbation theory (Kato, 1995) yields

$$\|DA(x^*)\| = \rho(DA(x^*)) + O(\varepsilon).$$

□

This justifies the eigenvalue-scaling argument used in Proposition 7.3.

## A.5 Identification of the Curvature Sensitivity Parameter

The parameter  $\gamma$  is estimable from organizational data.

Let

$$(\hat{R}_i, \hat{A}_i)$$

denote observed curvature-intensity and amplification measurements from organization  $i$ .

Define the least-squares criterion

$$Q(\gamma) = \sum_{i=1}^N \left( \hat{A}_i - e^{-\gamma \hat{R}_i} \right)^2.$$

**Definition A.7** (Curvature-sensitivity estimator).

$$\hat{\gamma} = \arg \min_{\gamma > 0} Q(\gamma).$$

**Theorem A.8** (Consistency). *Suppose*

$$\hat{A}_i = e^{-\gamma_0 \hat{R}_i} + \varepsilon_i,$$

*with*

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) < \infty.$$

*Then*

$$\hat{\gamma} \rightarrow \gamma_0 \quad \text{in probability as } N \rightarrow \infty.$$

*Proof.* Standard nonlinear least-squares consistency. □

Thus every free parameter of the framework is empirically identifiable.